Workshop 3 — Local AI with Ollama

Report | FSTT Local AI Assistant

Environment: AlmaLinux; AMD EPYC 9454P (96 logical CPUs); 250 GiB RAM; CPU inference with Ollama. Models: Qwen3.5 4B and Gemma 3 4B.

Answers to the 15 report questions

Q1. What is Local AI?

Local AI runs machine-learning inference on infrastructure controlled by the user or organization, rather than sending every request to a hosted model provider.

Q2. Difference between a Cloud LLM and a Local LLM?

A cloud LLM runs on a provider's infrastructure and is accessed remotely. A local LLM runs on the user's own hardware, offering more control over data and deployment, but requiring local computing resources and maintenance.

Q3. What role does Ollama play?

Ollama downloads, manages, loads, and serves local models. It exposes HTTP endpoints, such as /api/generate and /api/chat, that applications can call.

Q4. What is model quantization?

Quantization represents model weights with fewer bits, often reducing storage and RAM needs. It can improve inference efficiency, potentially at some cost in numerical precision or output quality.

Q5. Why does model size affect memory consumption?

A model with more parameters generally needs more memory to store its weights. Memory demand also depends on precision, context length, key-value cache, and runtime overhead.

Q6. Why can a smaller model sometimes be preferable?

Smaller models can load faster, use less memory, and respond with lower latency on limited hardware. They may be sufficient for focused tasks when their output quality is verified.

Q7. Advantages of local inference?

Local inference provides more data control, offline operation after installation, customizable deployment, and no per-token charge to a hosted LLM provider. Hardware and operating costs still apply.

Q8. Disadvantages of local inference?

Local deployment requires hardware, installation, maintenance, model updates, and security controls. CPU-only execution and limited memory can constrain model size, throughput, and concurrent users.

Q9. Explain the architecture of your Python application.

The Flask front end accepts questions. Python retrieves relevant local documents, combines the context with limited conversation history, and sends structured messages to Ollama's /api/chat endpoint. The application returns an answer, source filenames, and inference metrics; optional JSON schema mode structures selected responses.

Q10. Explain the role of the REST API.

The REST API connects Python to the Ollama runtime over local HTTP. A request carries a model name, prompt or messages, and inference options; the JSON response contains generated text and timing/token metadata.

Q11. What is RAG?

Retrieval-Augmented Generation first selects relevant information from an external knowledge base, then supplies it as context to the LLM. In this workshop, a lightweight keyword-based retriever selects local text documents before answer generation.

Q12. Difference between an LLM and a multimodal LLM?

A text-only LLM processes textual inputs; a multimodal LLM can additionally interpret other modalities, such as images. Gemma 3 4B was used as the vision-capable model in the optional activity.

Q13. Compare two local models experimentally.

On the reported single-run CPU benchmark, Qwen3.5 4B took 34.80 s and produced 811 tokens at 25.26 tokens/s, while Gemma 3 4B took 17.47 s and produced 822 tokens at 55.49 tokens/s. Their loaded sizes were 3.15 GB and 2.89 GB respectively. Gemma was faster in this run; repeated trials and independent quality scoring would be needed for a general conclusion.

Measured metric	Qwen3.5 4B	Gemma 3 4B
Disk size (GB)	3.32	3.35
Loaded size (GB)	3.15	2.89
Load time (s)	2.56	2.58
Response time (s)	34.80	17.47
Output tokens	811	822
Tokens/s	25.26	55.49
VRAM (GB)	0	0
System RAM used (%)	41.6	40.8
Quality (earlier coding task)	3/5	5/5

Results are from the supplied Part 16 screenshot. Quality scores refer to the earlier coding task, not the reinforcement-learning benchmark. Single run only.

Q14. Which model would you deploy on a computer with limited RAM? Explain.

Among the two tested configurations, Gemma 3 4B is the more memory-efficient option because its measured loaded size was 2.89 GB versus 3.15 GB for Qwen3.5 4B. On a much smaller machine, I would evaluate a smaller quantized model as well.

Q15. What security risks exist when exposing a local LLM server to a network?

Without a protected gateway, exposed endpoints can be abused for unauthorized inference, resource exhaustion, or access to sensitive prompts and outputs. Retrieved documents can carry prompt-injection instructions. Keep Ollama bound to localhost, use authentication and TLS at a reverse proxy, restrict firewall access, and apply resource limits and input/document controls.

Conclusion

The workshop demonstrates a fully local assistant combining Ollama, Python, Flask, conversation history, document retrieval, and structured responses. In the reported CPU benchmark, Gemma 3 4B provided higher generation throughput and lower latency, but repeated trials and separate accuracy checks are needed before deployment decisions.